SPE-229128-MS

Next Gen Carbon Capture Using Energy Efficient Simulated Moving Bed Concept

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

To meet their net zero goals for carbon emissions, industries are exploring clean and economically viable solutions for de-carbonization of the industrial processes. At present, liquid amine-based technologies are mostly used for carbon capture, however, these technologies face challenges in terms of high cost of CO_2 capture ($ 40-$50/ton of CO_2), toxic nature of solvents, solvent degradation, losses and its disposal. Considering the above challenges, adsorption-based processes utilizing solid sorbent materials are being explored as a cleaner and economical alternative for CO_2 capture from industrial flue gases and even for its direct air capture.

In the current work we introduce a novel energy efficient adsorption process leveraging upon advancement in process automation. Present concept mimics ideal counter current gas solid moving bed operation without moving the solid sorbent particles. Being counter current operation with stationary solids, the concept offers a higher gas to solid mass transfer rates to give higher CO_2 separation efficiencies, a process with zero solid pumping cost and free from any solid attrition. Further, unique inherent heat recovery and utilization feature of the process in temperature swing adsorption (TSA) mode leads to low sorbent energy requirements which makes the process highly economically viable.

To check the performance and efficacy of the concept, an adsorption simulation (ADSIM) model was developed for the process and simulation studies were performed for CO_2 capture from refinery flue gases containing 12-15 % CO_2 by volume. Results showed more than 90 % CO_2 recovery at 95 % CO_2 purity for proprietary sorbent material. Further, in temperature swing adsorption (TSA) mode energy requirement for the process estimated to be 0.265 MJ/ kg of CO_2 which was found to be 10-15 times lower than solvent regeneration energy required for state of art liquid amine-based technologies. The low sorbent regeneration energy for proposed process is envisaged to bring the CO_2 capture cost to $15-$20/ton of CO_2 which makes the process highly economically viable. Currently, 1 ton per day (TPD) plant is set up in India and concept is under pilot trials at TRL 5.

Keywords: Net zero, CO_2 capture, counter current adsorption, adsorption simulation (ADSIM), CO_2 purity, regeneration energy

Introduction

Liquid amine-based CO_2 absorption technologies are commercially proven for post combustion capture of CO_2 from industrial flue gases [Ziobrowski et al., 2022], however, their high capital expenditure, high solvent regeneration energy requirements, high cost of CO_2 capture coupled with solvent handling issues e.g. solvent toxicity, degradation and losses led to the exploration of new cleaner technologies [Zhao et al., 2011 and Budzianowski, 2016]. Study by Global CCS Institute (GCCSI) for CO_2 capture from coal power plants using 30 % (w/w) mono-ethanol-amine (MEA) solvent revealed that for coal power plants emitting 0.18 to 1.8 metric tonne CO2/year, CO_2 capture costs lies in the range of US$50–US$65 per tonne CO2 with costs generally decreasing for larger plants [Barlo H. and Shahi S.S.M., 2024]. Even though the most energy efficient state-of-art solvent based technologies e.g. CDRMax technology by Carbon Clean, UK reduced the capital investment by 30 % and solvent regeneration energy to 2.8 GJ/ton of CO_2 from 4 GJ/ton of CO_2 for conventional liquid amine based technologies, their high cost of CO_2 capture i.e. \$30-\$40/ton of CO_2 attracts attention to develop more clean and economical CO_2 capture technologies.

Owing to the high cost of CO_2 capture and solvent handling issue, CO_2 capture using solid adsorbent material has recently gained importance [Hack et al., 2022]. Such adsorption processes contrary to gas liquid contact in liquid amine-based absorption technologies involve gas solid contact and work on the principle of physisorption or chemisorption of CO_2 from gaseous mixture onto the solid adsorbent surface. A number of solid adsorbents e.g. amine-grafted silicas, metal organic frameworks (MOFs), activated carbon, activated alumina, zeolite-13X, supported alkali metal carbonates and hydrated carbonates, covalent organic framework (COFs) etc. have been employed in various works reported in the literature for CO_2 capture [Zou et al., 2001, Paradakhti et al., 2019 and Raganati et al., 2021] while fixed bed, fluidized bed and moving bed reactor concepts using either pressure swing or vacuum swing or temperature swing mode of solid regeneration are employed for CO_2 capture from various CO_2 sources [Jha et al., 2021]. Counter-current moving bed operations including counter-current flow of gas and solid streams offer higher mass transfer driving forces between gas and solid and thus produce the best CO_2 separation efficiencies. Kawasaki KCC process and Knaebal MBTSA process are examples of counter current moving bed processes. Direct steam contact for solid regeneration in KCC process demand highly hydrothermal stable adsorbent material. MBTSA process avoids direct steam contact by using hot flue gas for solid regeneration, however, main uncertainty of the process is heat transfer efficiency due to indirect contact between hot flue gas and solid material. Similarly, fluidized bed processes improve the gas solid contact and, therefore, the mass transfer, however, these processes need continuous pumping of the solid material which adds to the energy requirements. Another challenge with such processes is attrition and erosion of solid particles which pose solid handling and scale up challenges [Jha et al., 2021]. Also fluidized processes pose heat management challenges leading to inferior thermal efficiencies.

Considering limitations of the above processes, we are motivated to develop a unique adsorption concept which mimics ideal gas solid counter current moving bed process but only includes the advantages of counter current moving bed process while discarding the disadvantages e.g. solid movement and associated attrition challenges of moving beds including fluidized bed processes. We propose a concept where solid particles remain stationary, however, exposed to gas phase in a counter current manner and which involves direct contact between solid material and hot desorbent gaseous stream for solid regeneration. Concept involves equally dividing the adsorbent material in number of smaller fixed adsorbent beds. These beds are brought in contact with the gaseous feed, recycled extract stream (CO_2 rich stream), hot desorbent stream and cold recycled raffinate stream (CO_2 lean stream) in a sequence one after the other leading to all beds undergoing four basic operations of adsorption, enriching, desorption and pre-saturation sequentially in a cyclic process. This is facilitated by switching the injection of these streams from one bed to the other in a time-programmed manner. Switching is performed in such a manner that it mimics gas solid counter current moving bed operation. Contrary to conventional vacuum/pressure swing adsorption (VSA/PSA) processes which involve only adsorption and desorption steps for gas separation and where the separation efficiency is limited by the low partial pressure or concentration of the solute or adsorbate gas in the gaseous feed, proposed concept introduces two more steps of enriching and pre-saturation. In enriching, solid adsorbent material is exposed to higher partial pressure or concentration of solute gas, while in pre-saturation it is exposed to CO_2 lean raffinate. This improves the uptake of solute gas by adsorbent material and further dilutes the raffinate stream in solute to enhance the purity and recovery of extract and raffinate streams.

Figure 1 elaborates the movement of gas and solid in an ideal gas solid counter current moving bed operation. Solid particles move counter currently to gas stream and undergo pre-saturation, adsorption, enriching and desorption operations in one complete cycle. While in a pre-saturation zone, regenerated solid particles are contacted with CO_2 lean raffinate stream to capture the traces of CO_2. Pre-saturation zone also acts as cooling step for the solids from desorption zone in temperature swing adsorption (TSA) mode. CO_2 from the feed is adsorbed in adsorption zone. Post adsorption, solid enters an enriching zone where it is brought in contact with high CO_2 partial pressure recycled gas stream from desorption zone. Post enriching, finally the CO_2 loaded solids are regenerated in desorption zone. In the proposed concept, as stated earlier, solid particles or beds remain stationary, however, counter current operation between gas and solid is achieved by continuously and simultaneously switching the feed injection and separated extract (CO_2 rich stream) and raffinate (CO_2 lean stream) streams ejections from one solid adsorbent bed to another as shown in Figure 2. The advantage of such a process is that it is free from solid pumping and attrition challenges and at the same time offer CO_2 separation efficiencies close to ideal counter current true moving bed operations. Further, in Temperature Swing Adsorption (TSA) mode of solid regeneration, proposed concept facilitates effective recovery of heat supplied to adsorbent bed for its utilization for solid regeneration in subsequent cycles. Figure 3 represents heat management and integration for the proposed concept. Thus, the present work proposes a Simulated Moving Bed (SMB) process with continuous switching of feed and other gaseous streams injection/ejection from one solid bed to the other without moving the solids and which offers excellent heat recovery and utilization facilitated by switching of the valves.

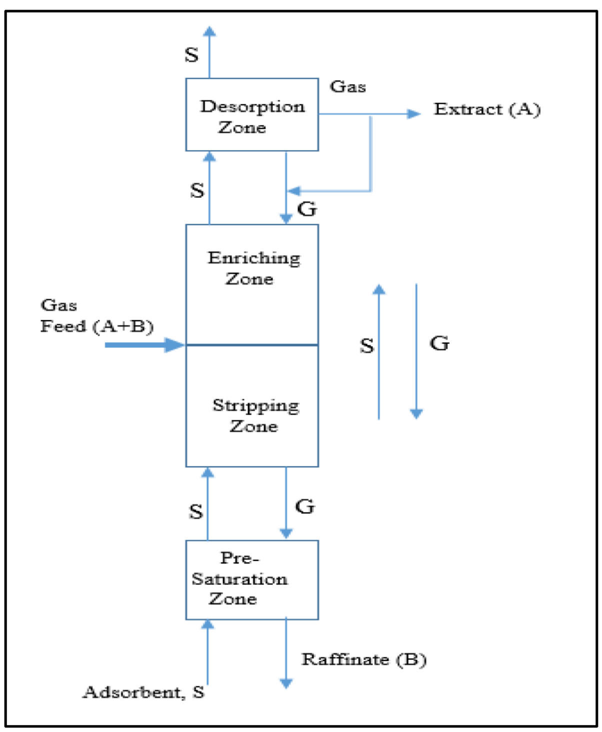

Figure 1—Counter-current gas solid true moving bed (TMB) operation

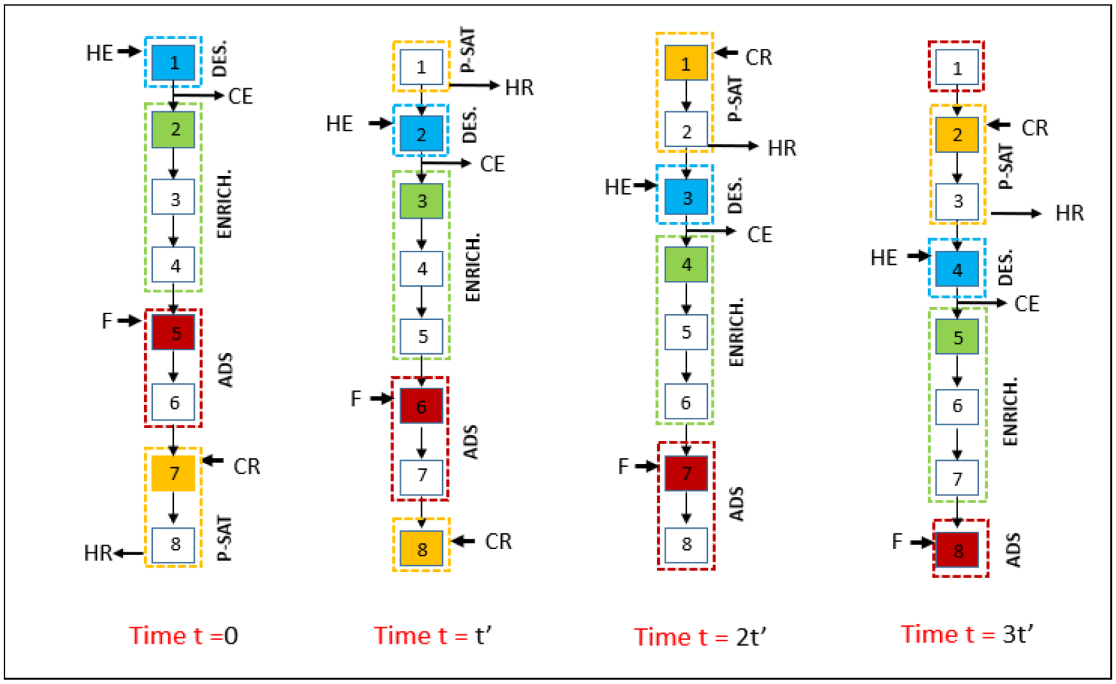

Figure 2—Switching of feed (F) and product streams e.g. hot extract (HE), cold extract (CE), hot raffinate (HR) and cold raffinate (CR) in the proposed concept

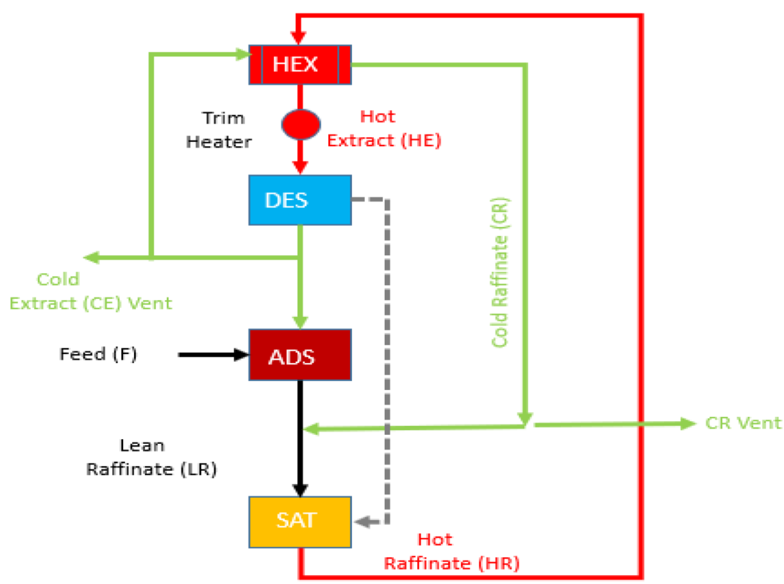

Figure 3—Heat management in proposed SMB concept; HEX (Heat Exchanger), DES (Desorption zone), ADS (Adsorption zone), SAT (Pre-saturation zone)

Methodology

Development of SMB-TSA ADSIM Customized Model

To check the efficacy of the proposed concept, an eight bed Simulated-Moving-Bed-Temperature-Swing-Adsorption (SMB-TSA) customized model developed in Aspen ADSIM, a process simulation software for adsorption operations, for CO_2 capture from a gas mixture containing 85 % (V/V) N_2 and 15 % (V/V) CO_2. Figure 4 provides the flow sheet for the SMB-TSA process developed in ADSIM.

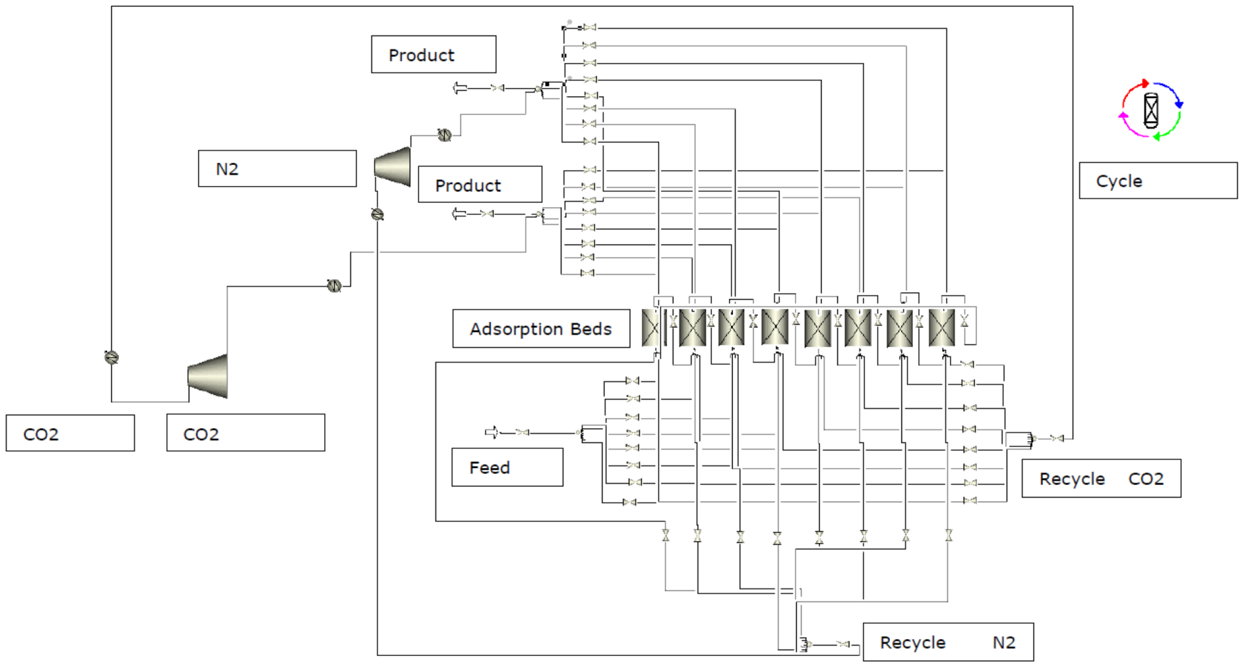

Figure 4—Flow sheet of eight-bed simulated moving bed (SMB) system

In the simulation process, feed is injected into solid adsorbent beds while injection is programmed and controlled by opening and closing of switch or ON/OFF valves. CO_2 adsorbed bed is then moved to enriching zone by virtue of switching the valves where it is contacted with recycled CO_2 rich stream from desorption zone with higher CO_2 partial pressure or concentration for maximization of solid CO_2 uptake. Post enriching, bed is moved to desorption zone, again by virtue of switching valves, where adsorbed CO_2 is knocked out by using hot CO_2 stream. Part of regenerated CO_2 is recycled to downstream solid beds while part of it goes to recovery. Hot raffinate (N_2 rich stream) from pre-saturation zone and cold extract (CO_2 rich) stream from desorption zone exchange heat in a plate type heat exchanger. Plate type heat exchanger with high heat transfer area ensures effective transfer of heat from hot raffinate stream to cold extract stream. This heat exchange increases the temperature of cold extract stream to make it a hot stream whose temperature is further adjusted to desorption temperature in a trim heater before it contacts the solid under desorption. For simulation studies, eight beds were distributed equally among adsorption, enriching, desorption and pre-saturation zones. All above operations are achieved simultaneously by careful sequencing of opening and closing of valves and are enabled through a "cycle organizer".

Models Selection

Thermodynamic Model. In the present work, since binary gaseous mixture of CO_2 and N_2 is used as feed to demonstrate the efficacy of the proposed concept for CO_2 capture, an ideal gas model is chosen to estimate the thermodynamic properties and to represent the behaviour of these two feed gases CO_2 and N_2. The number of components can be added to the model and the properties prediction model can be changed with Aspen Plus's property package accessible directly from the Aspen Plus Adsorption environment. For every component added, pure component adsorption Isotherm for the component will be required while configuring the adsorption beds.

Gas Flow Hydrodynamic Model. Reactor system being a micro-reactor system, a simplified one-dimensional (1-D) axial model assuming variation of concentration, temperature and pressure only along the length of the bed and no variation in radial direction chosen to represent the flow dynamics of gases in the bed. The simulation incorporates non-isothermal conditions with gas-based convection as the primary mode of heat transfer. Similarly, mass transfer relies primarily on convection for its transport dynamics. Pressure drop calculations are based on the Ergun equation defined by equation (1).

$$\frac{\Delta P}{L} = 150(1-\varepsilon)^2\mu\frac{u}{\varepsilon^3 dp^2} + 1.75\frac{(1-\varepsilon)\rho u^2}{\varepsilon^3 dp} \tag{1}$$

where, ΔP is pressure drop along the length of the bed, L is length of the packed bed, μ is dynamic viscosity of the fluid, ρ is density of the fluid, u is superficial velocity of the fluid, ε is bed porosity and do is equivalent particle diameter.

Adsorption Isotherm Model. To represent adsorption behaviour of CO_2 and N_2 on adsorbent (zeolite 13 X) temperature dependent Extended Langmuir 2 concentration-based adsorption isotherm model was selected. Equation (2) gives Extended Langmuir 2 concentration-based adsorption isotherm model. Owing to the exothermic nature of adsorption process and desorption being performed by heating the adsorbent material, a temperature dependent adsorption isotherm model was selected to capture the adsorption and desorption behaviour more closely to the reality.

$$\omega_i = IP_{1i}e^{IP_{2i}/Ts}C_i / \left\{1 + \sum_k \left(IP_{3k}e^{IP_{4k}/Ts}C_k\right)\right\} \tag{2}$$

where, ω_i is equilibrium loading of component 'i', IP1, IP2, IP3, IP4 are isotherm parameters, Ts is solid phase temperature, C is concentration of gaseous components.

Adsorbent Material Selection

Efficacy of adsorption process depends not only on the ability of the material to readily and selectively adsorb solute gas from gaseous mixture but also to facilitate fast desorption from its adsorption sites. Zeolite 13X is a highly porous crystalline material which offers large surface area for adsorption of CO_2. Further, presence of cations typically Na^+ due to electrostatic interactions with CO_2 enhance affinity towards CO_2 and, therefore, CO_2 adsorption capacity of zeolite 13X. For this reason, CO_2 uptake capacities (5-6 mmoles/g) of zeolite 13X are higher than many of the adsorbents in the business. Hence, zeolite-13X was chosen as adsorbent material for separation of CO_2 from binary mixture of CO_2 and N_2. Pure component isotherm data for both CO_2 and N_2 was taken from literature [Ko et al., 2005]. Extended Langmuir Adsorption Isotherm Model as described above was chosen to represent the adsorption of CO_2 and N_2 on zeolite-13X. Adsorbent properties e.g. absolute density, bulk density, bed inter-particle voidage, intra-particle voidage, particle diameter, mass transfer coefficients, heat transfer coefficient, thermal conductivity, specific surface area of adsorbent were input to the simulation model to capture adsorbent behavior.

Simulation Studies

Pure component isotherm data, binary break-through studies data for CO_2/N_2 gas mixture and physical properties of proprietary adsorbent material were input to the simulation model. Simulation studies performed to enrich a binary feed mixture comprising 85 % (V/V) N_2 and 15 % (V/V) CO_2 to produce CO_2 rich extract phase and CO_2 lean raffinate phase. All simulation runs were carried out at 30 ˚C adsorption temperature while desorption was carried out at 150 ˚C. As solid particles undergo four different operations or zones e.g. adsorption, enriching, desorption and pre-saturation in SMB process, in simulation studies, two solid beds at a time were considered under each zone and thus total eight beds were divided into 2-2-2-2 configuration, though depending upon the number of solid adsorbent beds under each zone, process can be performed for different configurations also. Effects of feed flow rate, port switch time i.e. feed and product streams switching time from one bed to another, and number of cycles after achieving cyclic steady state performed to find out the CO_2 purity and recovery in the process. Energy requirement studies performed in Aspen HYSYS simulation software. Desorption of CO_2 from solid adsorbent surface was carried out at

150 ˚C using part of CO_2 rich extract phase as desorbent gas. Desorption temperature of 'desorbent CO_2' is achieved by facilitating heat exchange between desorbent CO_2 and hot raffinate stream (N_2 rich) in a plate type heat exchanger and further heating the desorbent CO_2 in a trim heater to bring its temperature to desorption temperature of 150 ˚C.

Results & Discussions

Effect of Feed Flow Rate

To study the effect of feed flow rate, simulation runs were carried out at different feed flow rates e.g. 14, 28, 42, 200 and 250 SLPM (standard liter per minute) keeping port switch time (200 sec) and no. of cycles (02) constants after achieving cyclic steady state. Figure 5 represents the effect of feed flow rate on CO_2 purity (i.e. vol % of CO_2 in extract phase) and CO_2 recovery (CO_2 in raffinate as wt. % of CO_2 in feed). A decreasing trend in CO_2 purity and recovery was observed on increasing the feed flow rate which can be attributed to reduced contact time between gas and solid phases at higher flow rates resulting in incomplete CO_2 adsorption and thus its slip into raffinate stream. On the contrary, N_2 adsorption is favored over CO_2 at high flow rates owing to higher concentration of N_2 in the feed. As revealed by simulation studies, a CO_2 purity more than 99 % and CO_2 recovery of more than 95 % could be achieved for zeolite-13X adsorbent at feed flow rate of 14 SLPM.

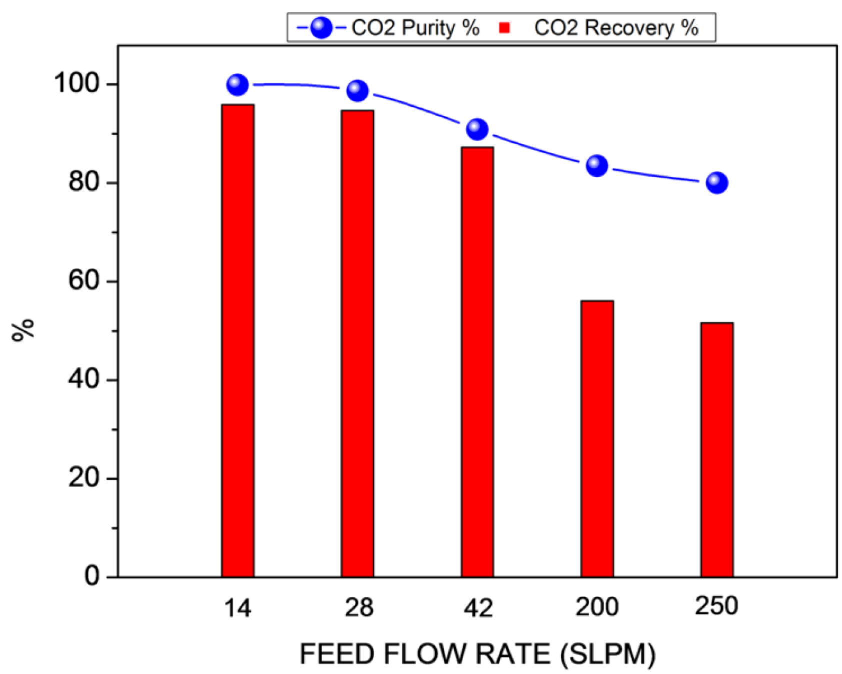

Figure 5—Changes in CO2 purity and recovery with feed flow rate

Effect of Port Switch Time (PST)

Next to the study of feed flow rate, effect of port switch time (PST) i.e. valve switching time studied over CO_2 purity and recovery. Study was performed at constant feed flow rate 14 SLPM and 02 no. of cycles after achieving cyclic steady state. It was observed that increasing the port switch time (PST) has a negative effect on CO_2 purity and recovery (Figure 6). On reducing the port switch time from 1000 sec to 200 sec, CO_2 purity and recovery increased from 52.87 % and 50.76 % to 99.91 % and 95.91 % respectively. Such an increase in CO_2 purity and recovery can be attributed to lesser exposure of low CO_2 concentration feed to solid adsorbent giving the solid in enriching zone opportunity to pick up more CO_2 from higher CO_2

concentration recycled gas from desorption zone. Higher CO_2 concentration recycled gas pushes more CO_2 adsorption on the solid surface due to higher mass transfer driving forces. Hence, SMB operation with port switch time (PST) as low as practically possible should be practiced.

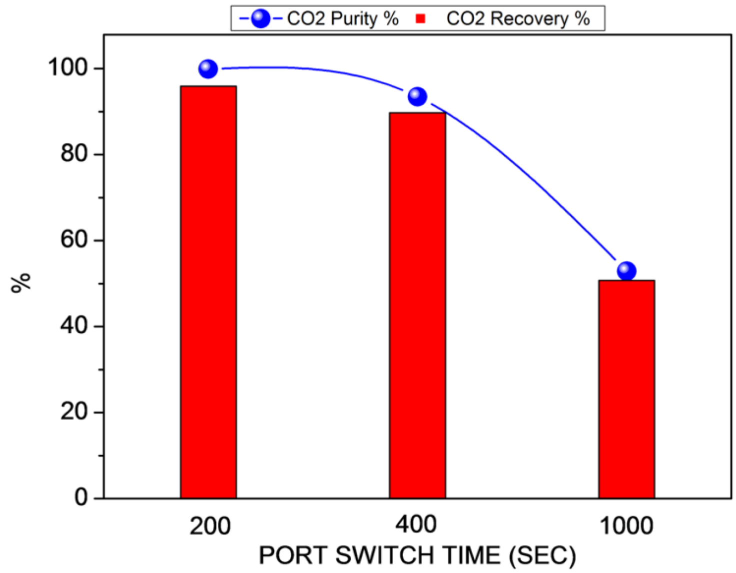

Figure 6—Changes in CO2 purity and recovery with port switch time (PST)

Effect of No. of Cycles

Effect of number of cycles studied to judge the changes in CO_2 purity and recovery on performing the SMB runs for longer duration. The simulation study was performed at constant feed flow rate 14 SLPM and constant port switch time (PST) 200 sec while number of cycles after achieving study state varied from 2 to 20. Figure 7 gives changes in CO_2 purity and recovery with respect to the number of cycles. No change in CO_2 purity and recovery observed over 20 no. of cycles, which signified stable performance of adsorbent material.

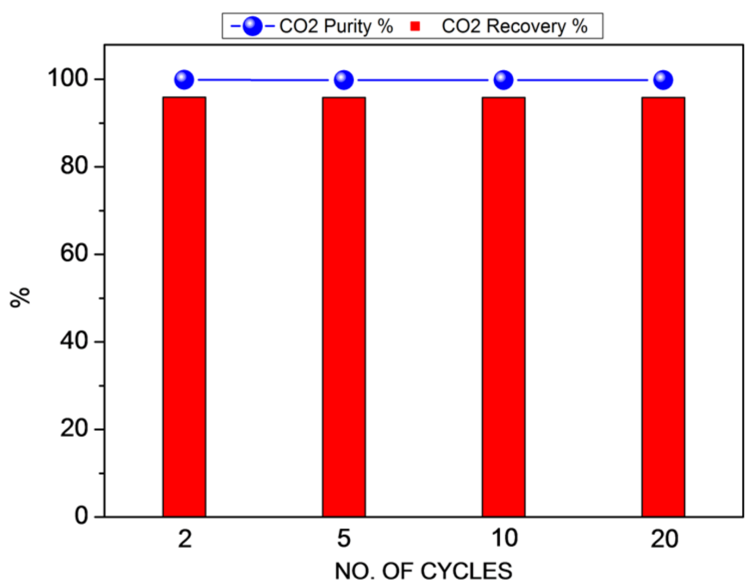

Figure 7—Changes in CO2 purity and recovery with number of cycles

Temperature Profile of Solid Adsorbent Bed

Figure 8 gives the temperature profile of adsorbent bed in each zone. Figure 7(a) reveals temperature profile in adsorption zone for first 200 sec. Temperature profile reveals isothermal adsorption at 30 ˚C. By virtue of switching valves, bed under adsorption zone moves to enriching zone after 200 sec of run, Figure 7(b) reveals rise in temperature of bed from 30 ˚C at 200 sec to close to 40 ˚C at 600 sec at the end of enriching. Figure 7(c) reveals an increase in bed temperature to desorption temperature of 150 ˚C from 600 sec to 650 sec and temperature remains at 150 ˚C till completion of desorption operation. Figure 7(d) shows cooling of the adsorbent bed from 150 ˚C to 30 ˚C in pre-saturation zone in run time from 1000 sec to 1200 sec using cold raffinate stream (N_2 rich stream). Hence, in a complete cycle, a solid bed temperature changes from 30 ˚C in adsorption zone to 40 ˚C in an enriching zone and from 40 ˚C in enriching zone to 150 ˚C in desorption zone and from 150 ˚C in desorption zone to 30 ˚C in pre-saturation zone for subsequent adsorption.

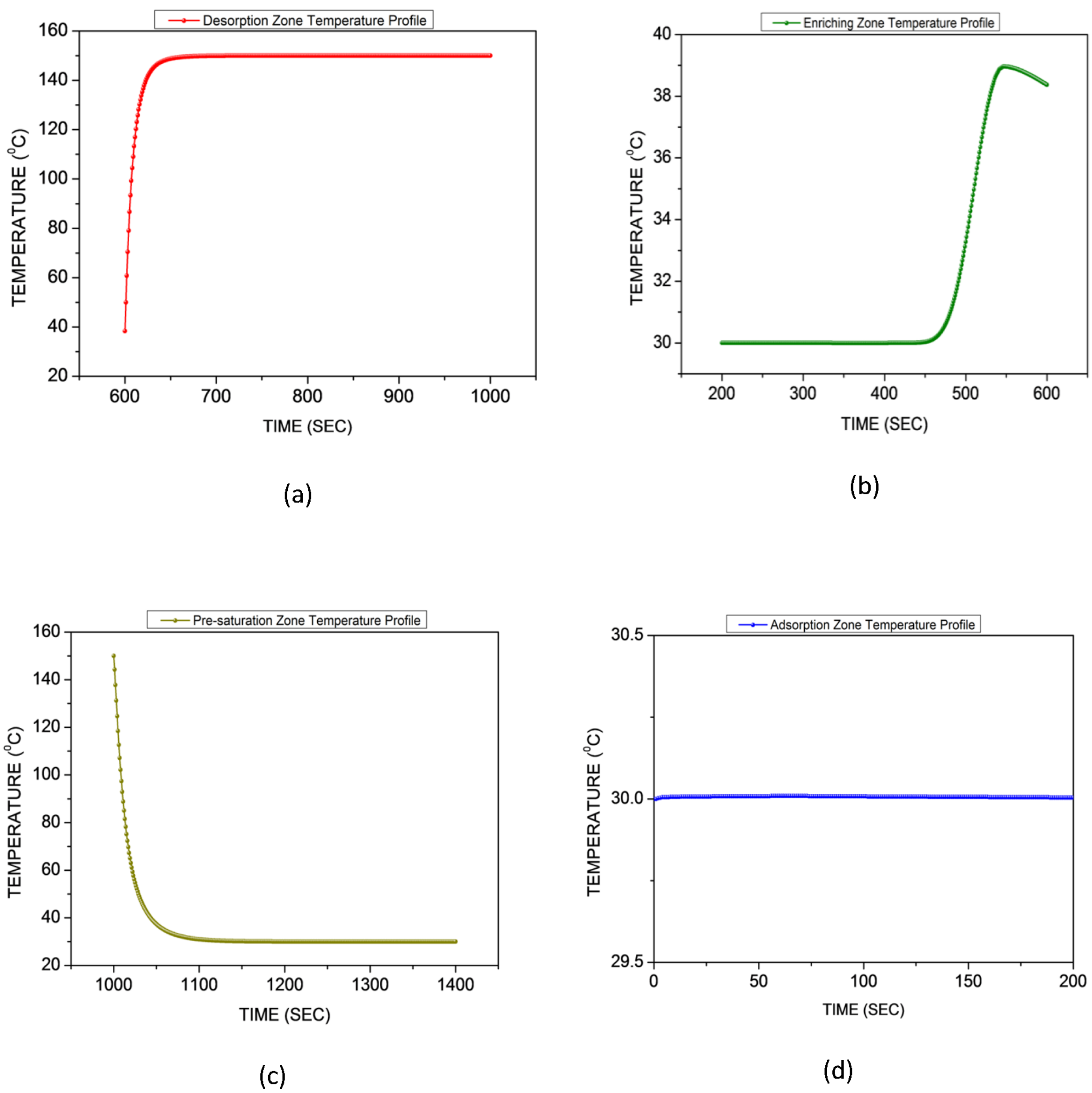

Figure 8—Temperature profile of solid adsorbent bed in each zone; (a) adsorption zone, (b) enriching zone, (c) desorption zone, (d) pre-saturation zone

Energy Calculations

Energy calculations for estimating energy requirements per kg of CO_2 captured were performed for 14 SLPM feed flow rate, 200 sec port switch time and 02 nos. of cycles. Adsorption of feed over zeolite-13X was carried out at 30 ˚C while desorption was carried out at 150 ˚C using part of CO_2 recovered. Energy calculations were performed using plate type heat exchanger with high heat transfer area in Aspen HYSYS process simulation software. Hot raffinate stream from pre-saturation zone is brought in contact with cold extract stream from desorption zone by means of plate and fin type of heat exchanger. Plate and fin type of heat exchanger was chosen to offer higher heat transfer area (~ 1000 m^2/m^3) to achieve maximum heat transfer between hot raffinate and cold extract streams. Subsequently, hot extract stream is further heated in a trim heater (E-101) to provide additional heat required to bring the solid bed temperature to 150 ˚C in desorption zone. Figure 9 gives the heat transfer system used in the proposed SMB concept. From heat duty calculations of E-101, a total of 0.265 MJ energy per kg of CO_2 was required to carry out desorption of adsorbed CO_2. The estimated energy is found to be 10-15 times lower than solvent regeneration energy required (~ 2.8 MJ/Kg of CO_2) even by state of art solvent-based technologies.

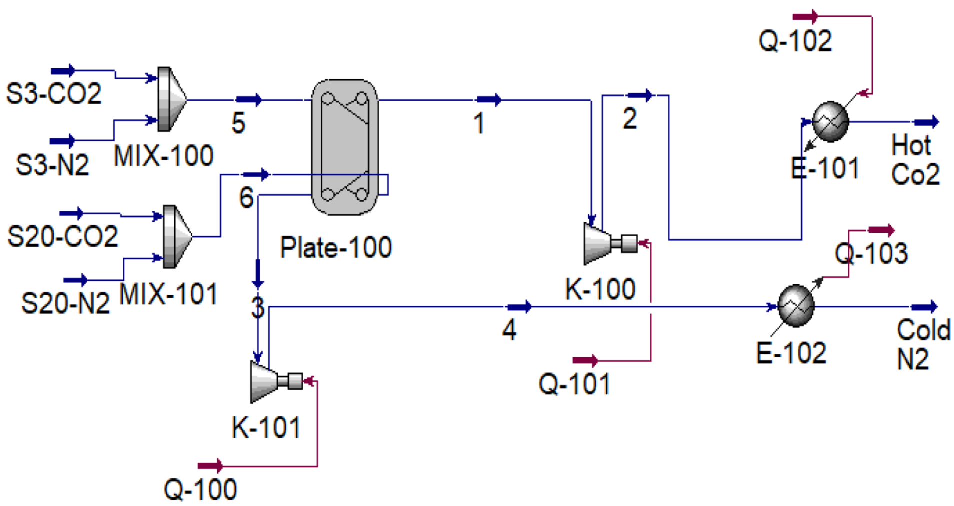

Figure 9—Heat exchange between hot raffinate (N_2 rich) and cold extract (CO_2 rich) streams using plate type heat exchanger

Conclusions

Simulation studies for proposed simulated moving bed (SMB) concept using zeolite-13X as adsorbent material revealed enriching of binary CO_2/N_2 gas mixture containing 15 % CO_2 (V/V) to CO_2 purity level of 99.91 % (V/V) in the extract phase with 95.91 % recovery based on the mass of CO_2 in the feed gases. A low feed flow rate and port switch time favored maximization of CO_2 purity and recovery. Excellent heat integration feature of the proposed concept resulted in solid regeneration energy requirements of 0.265 MJ/ kg of CO_2 which was 10-15 times lower than solvent regeneration energy requirements for state of art liquid amine-based technologies. Energy requirements can further be optimized by the proper choice of adsorbent material. However, such low regeneration energy is envisaged to bring down the CO_2 capture cost to \$15 - \$20/ton of CO_2.

Hence, the current simulation work over an eight-bed simulated moving bed system successfully demonstrated the efficacy of the proposed simulated moving bed (SMB) concept for CO_2 capture and provided future direction for pilot scale demonstration of the concept. Based on the outcome, 1 ton per day (TPD) pilot plant is set up in India. Currently, the concept is under pilot trials at TRL 4-5.

Acknowledgements

We would like to acknowledge BPCL R&D Management to approve the proposed concept, fund allocation and allow us to pursue research work on the proposed concept. Special acknowledgements to Dr. Sunil Saha, Director, ModeliCon, Bangalore to provide his support for development of Aspen Customized Model (ACM) for Simulated Moving Bed (SMB) simulation studies as per BPCL R&D concept.

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